

Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Box plot & normal distribution

Session 4

Box plot & normal distribution

Session 4

Objectives:

- Construct and interpret boxplots.
- Use boxplots to detect outliers.
- Understand the Normal distribution and its importance.
- Compute and interpret z-scores.

Percentiles

After measuring the central value, the spread, we can now estimate the relative position. For this we can divide the data into equal parts and state in which the value lies.

If you divide your data into **100 parts**; this is called **PERCENTILES**

- **The p-th percentile is the value below which p% of the data fall.**

Percentiles are everywhere: in exam reports, health charts (baby growth percentiles), or even in income statistics. They are one of the most practical measures of position in statistics

Ex: If your test score is at the 90th percentile, you did better than 90% of the class.

Compute Percentiles

Steps:

- **Order** the data from smallest to largest.
- Compute position: **$I = p/100 * N$** .

Then:

- If I is an integer, the percentile value is the data value at position
- If I is not an integer, use interpolation between the two closest ranks or take the maximum

Ex: Dataset: [10, 20, 30, 40, 50]

$N=5$

40th percentil -> $I = 0.4 * 5 = 2$ -> 20

45th percentil -> $I = 0.45 * 5 = 2.25$. Next larger number: 3. Therefore, 30

If you interpolate $(20+30)/2=25$

Quartiles

Quartiles are just special percentiles. You divide your data into **4 equal parts**

- Q1 is the 25th percentile: 25% of the data are less or equal to this value
- Q2 is the median : 50% of the data are less or equal to this value
- Q3 is the 75th percentile : 75% of the data are less or equal to this value

Percentages, Percentiles & Quartiles – What's the Difference?

Percentage (%):

- A way to express a portion of a whole (out of 100)
- Example: 60% of survey respondents chose Option A
- Used in Pie Chart

Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
- Used in standardized tests, income distribution, etc.

Quartiles

Special percentiles that divide data into four equal parts:

- Q1 (25th percentile) – 25% of values are below this
- Q2 (50th percentile) – the median
- Q3 (75th percentile) – 75% of values are below this
- Used to describe spread in boxplots

Percentages, Percentiles & Quartiles – What's the Difference?

Percentage (%):

- A way to express a portion of a whole (out of 100)
- Example: 60% of survey respondents chose Option A
- Used in Pie Chart

Ex: weights in the family: 25, 17, 32, 11, 40, 35, 13, 5, and 46.

Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
- Used in standardized tests, income distribution, etc.

- Calculate the percentages of people with a weights greater than the average
- Calculate the percentages of people with a weights greater than the median
- Calculate the lower and upper quartiles (Q1, Q3)
- Calculate the 90th percentile

Quartiles

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Percentages, Percentiles & Quartiles – What's the Difference?

Percentage (%):

- A way to express a portion of a whole (out of 100)
- Example: 60% of survey respondents chose Option A
- Used in Pie Chart

Ex: weights in the family: 25, 17, 32, 11, 40, 35, 13, 5, and 46.

Ordered: **5,11,13,17,25,32,35,40,46**

Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
- Used in standardized tests, income distribution, etc.

- Calculate the percentages of people with a weights greater than the average: **55%**

- Calculate the percentages of people with a weights greater than the median: **44%**

Quartiles

Special percentiles that divide data into four equal parts:

- Q1 (25th percentile) – 25% of values are below this
- Q2 (50th percentile) – the median
- Q3 (75th percentile) – 75% of values are below this
- Used to describe spread in boxplots

- Calculate the lower and upper quartiles of the following: **13,35**
- Calculate the 90th percentile: **46**

Five-number summary

The five-number summary is the foundation of the boxplot. It gives us five critical values:

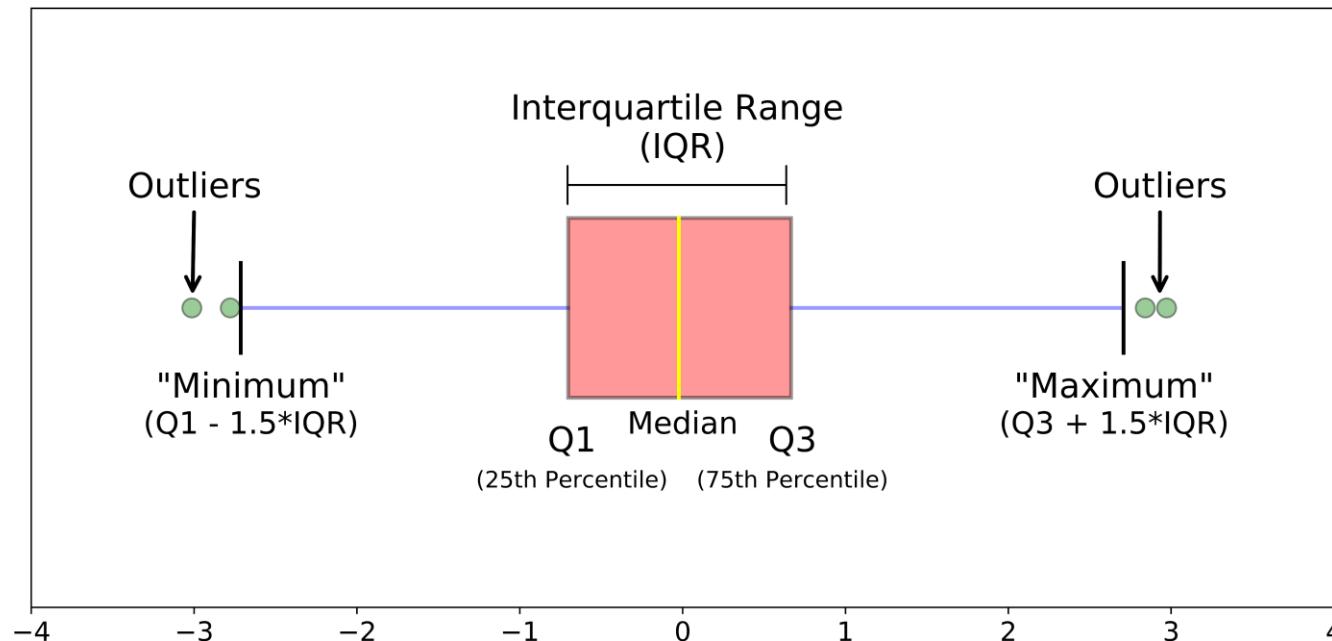
- min
- Q1
- Q2
- Q3
- max

This summary is very compact but powerful. With it, we can describe the dataset's range, central tendency, and variability, all in just five numbers.

Boxplot charts

A box plot uses boxes and lines to depict the distributions of one or more groups of numeric data. It is the graphical representation of the 5-number summary.

Your primary objective is to show distributions of numeric data values, especially when you want to compare them between multiple groups



Interquartile range

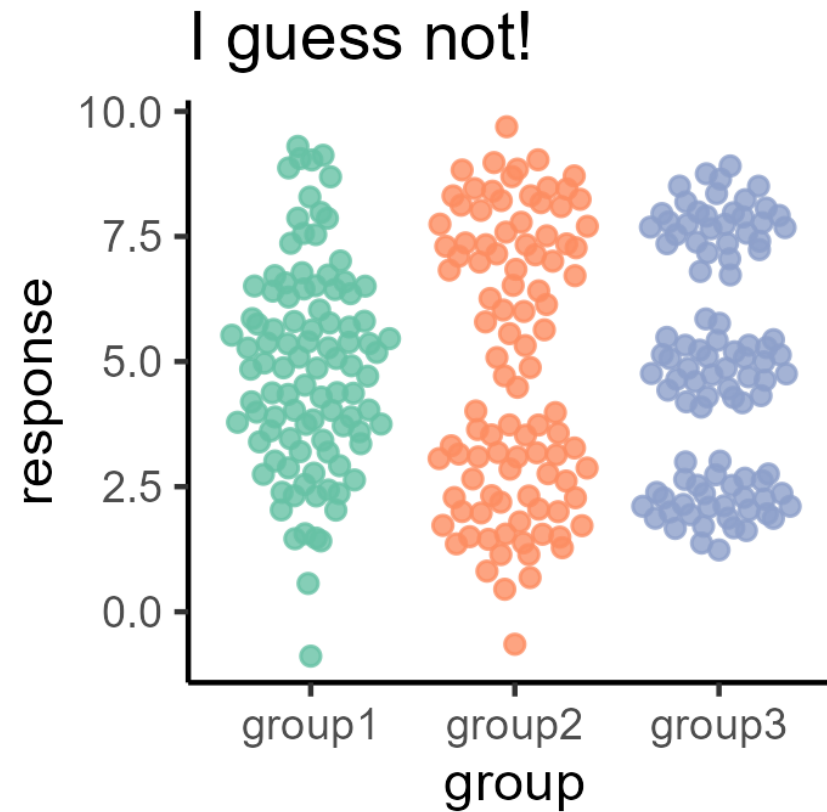
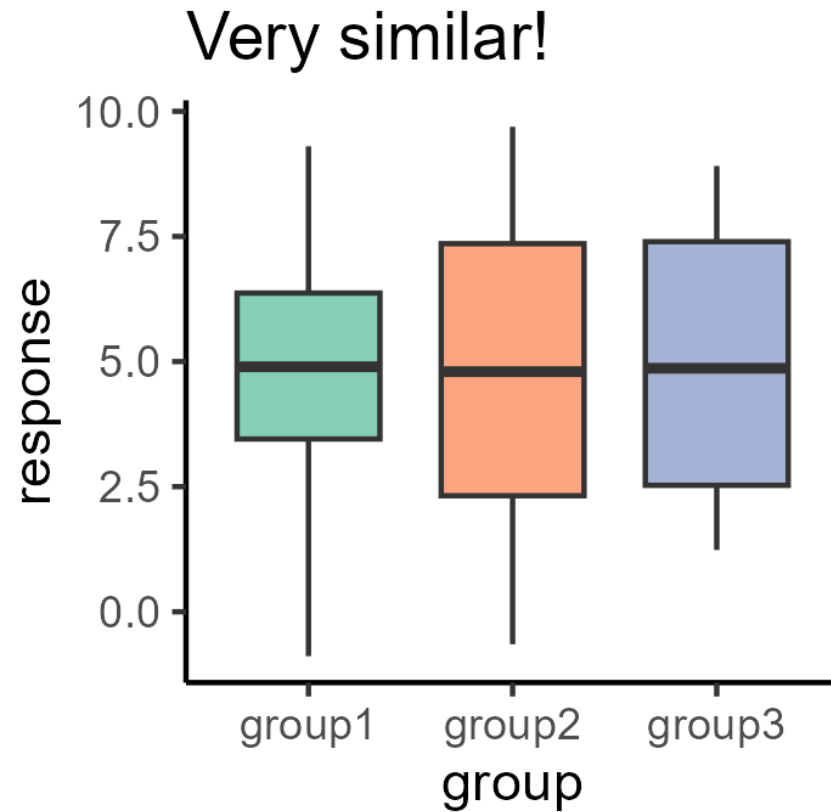
The interquartile range is the range of the middle 50% of the data given by:

$$\text{IQR} = Q3 - Q1$$

Where Q3 is the third quartile and Q1 the first quartile

Boxplot charts

If we are not comparing unimodal distributions, boxplots can be misleading and hide the real distributions. Use violin charts



Standard score

As we have seen, using percentiles (boxplot) to compare different population is not the best. For example, if one test has an average of 60 and another has an average of 80, percentiles don't help us compare across the two.

The z-score tells us how far a value is from the mean, expressed in units of standard deviation. It is a universal scale, which allows us to compare scores across different datasets.

$$Z = \frac{x - \mu}{\sigma}$$

Here:

- x is the value of interest from the population
- μ is the population mean and
- σ is the population standard deviation

VERY USEFUL TO IDENTIFY OUTLIERS!!!!!!!!!!

Standard score

Example:

- **Exam A:** $\mu=70$, $\sigma=10 \rightarrow$ **Student A scored 85**
- **Exam B:** $\mu=50$, $\sigma=5 \rightarrow$ **Student B scored 60**

Who did better on the exam?

Standard score

Example:

- **Exam A:** $\mu=70$, $\sigma=10 \rightarrow$ **Student A scored 85**
- **Exam B:** $\mu=50$, $\sigma=5 \rightarrow$ **Student B scored 60**

Student A $\rightarrow z = (85-70)/10 = 1.5$

Student B $\rightarrow z = (60-50)/5 = 2$

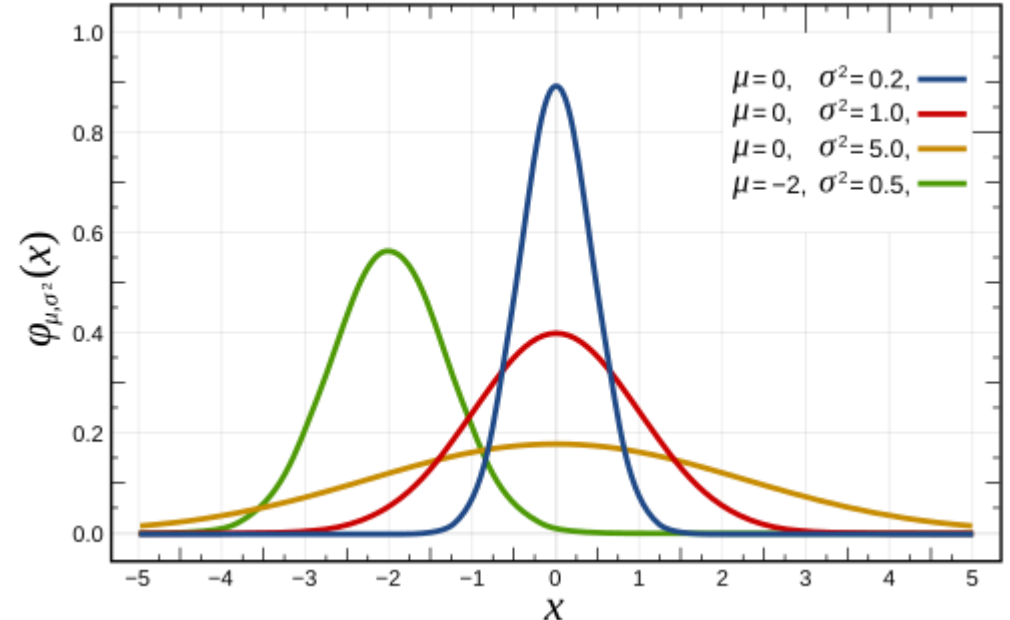
Student B performed better relative to their peers (2 vs 1.5)

Normal distribution

Now that we've seen z-scores, let's look at the distribution where they are most useful: the **normal distribution**, also known as the bell curve

- Symmetric, bell-shaped curve
- Defined by two parameters:
 - Mean $\mu \rightarrow$ center of the distribution
 - Standard deviation $\sigma \rightarrow$ spread of the distribution
- Notation: $X \sim N(\mu, \sigma^2)$
- Mean = Median = Mode
- $f(x)$ density probability:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Normal distribution and z-score

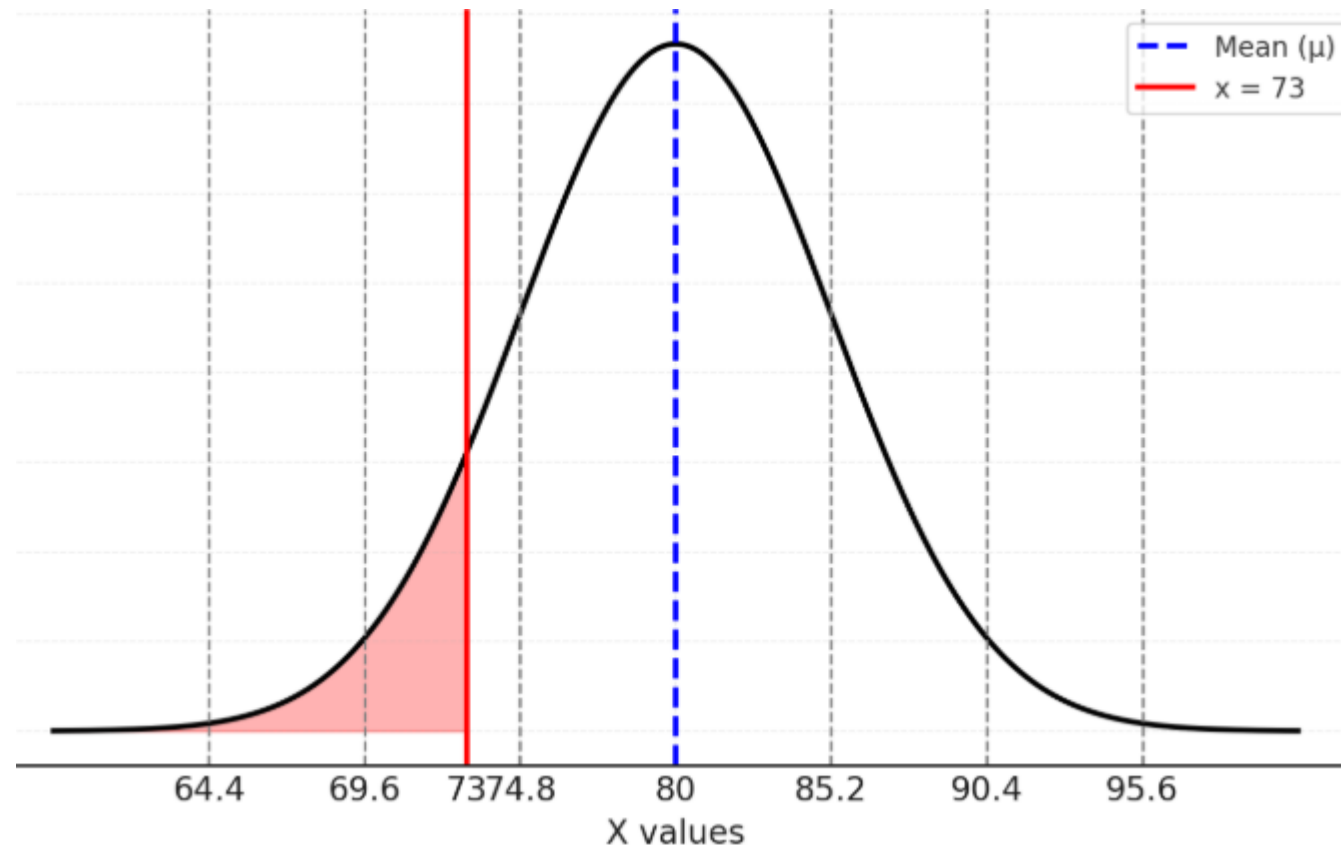
A normally distributed population of test scores has a mean of 80 and a standard deviation of 5.2.

Find the percentage of scores that lies below 73

Normal distribution and z-score

A normally distributed population of test scores has a mean of 80 and a standard deviation of 5.2.

Find the percentage of scores that lies below 73



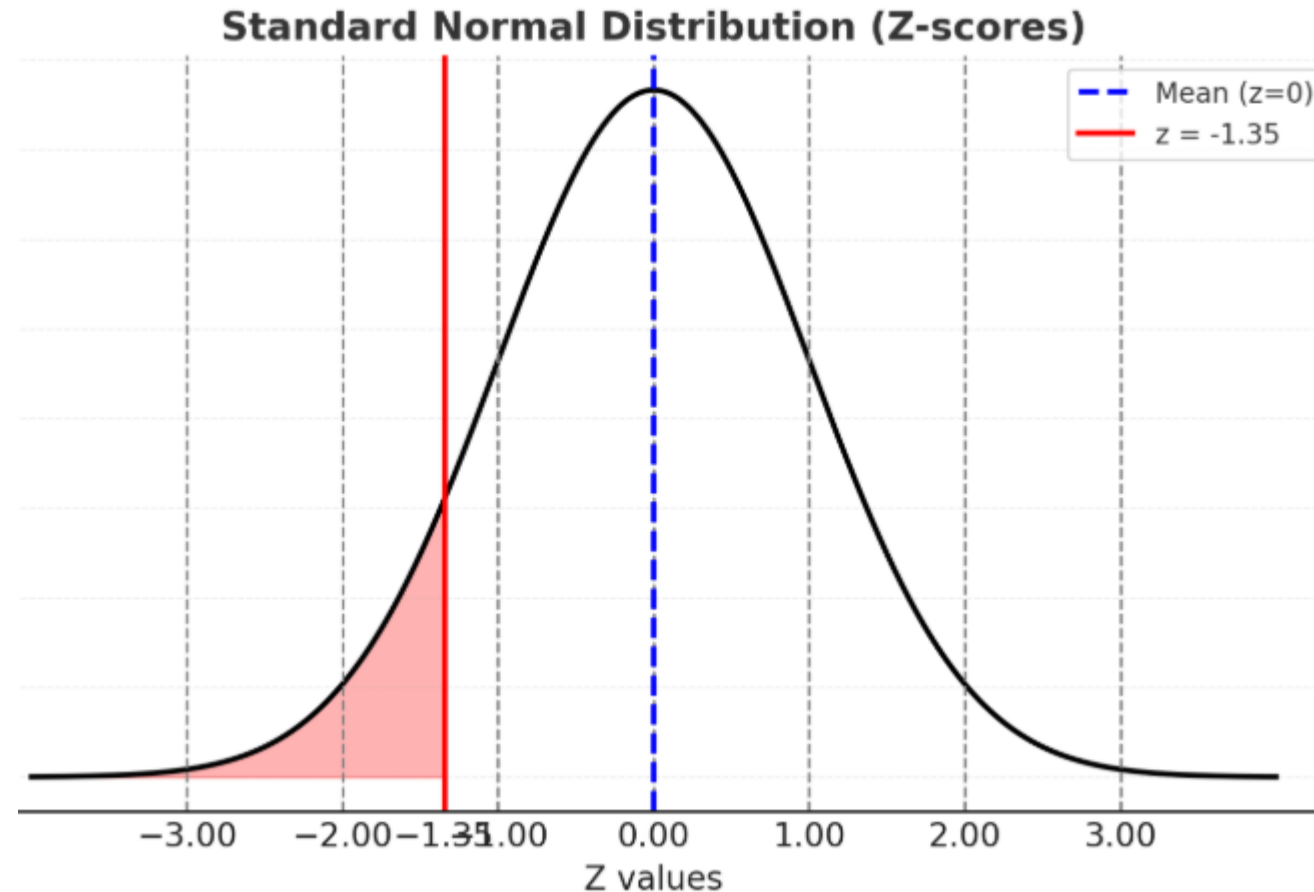
Normal distribution and z-score

- First you can calculate the z-score of 73

$$z\text{-score} = (73 - 80) / 5.2 = -1.34$$

Then you can normalize your distribution!!

$N(0,1)$



Normal distribution and z-score

Look at the zscore table

9.01%

Therefore, the percentage of score below 73 is 9.01%

1.1538 -> 0.87493

0.3846 -> 0.35569 -> 0.64431

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003

Exercices

Exercise 1:

Data (weights): **25, 17, 32, 11, 40, 35, 13, 5, 46**

- Find **Q1** and **Q3**
- Compute the **five-number summary** and the **IQR**.
- Give the **lower** and **upper fences** ($1.5 \times \text{IQR}$). Flag any outliers.

Exercices

Solution 1:

Sorted: 5, 11, 13, 17, 25, 32, 35, 40, 46 (N=9)

Q1: $I = 0.25 \cdot 9 = 2.25 \rightarrow$ pos 3 $\rightarrow$ **13**

Q3: $I = 0.75 \cdot 9 = 6.75 \rightarrow$ pos 7 $\rightarrow$ **35**

Five-number: **(5, 13, 25, 35, 46)** (min, Q1, Q2, Q3, max)

IQR = $35 - 13 =$ **22** $\rightarrow 1.5 \text{IQR} = 33$

Fences: **-20 (Q1-1.5IQR)** and **68 (Q3+1.5IQR)** $\rightarrow$ no outliers

Exercices

Exercise 2:

Two classes' exam summaries:

Class A: min=42, Q1=58, median=66, Q3=74, max=92

Class B: min=48, Q1=60, median=65, Q3=85, max=98

- Which class is **more variable**? Explain using IQR and range.
- Which class is **skewed**, and in what direction? Explain.
- For each class, estimate the **proportion above 75** using the boxplot logic.

Exercices

Solution 2:

- **IQR (Q3 – Q1):** A: $74 - 58 = 16$ and B: $85 - 60 = 25$
→ Class B has **more variability in the middle 50%** of data.
→ Both classes have the same **overall spread**, but B's middle spread is larger.
- Class A: Median (66) is roughly centered between Q1 (58) and Q3 (74). Whiskers: lower ($42 - 58 = 16$), upper ($74 - 92 = 18$). → Almost symmetric.

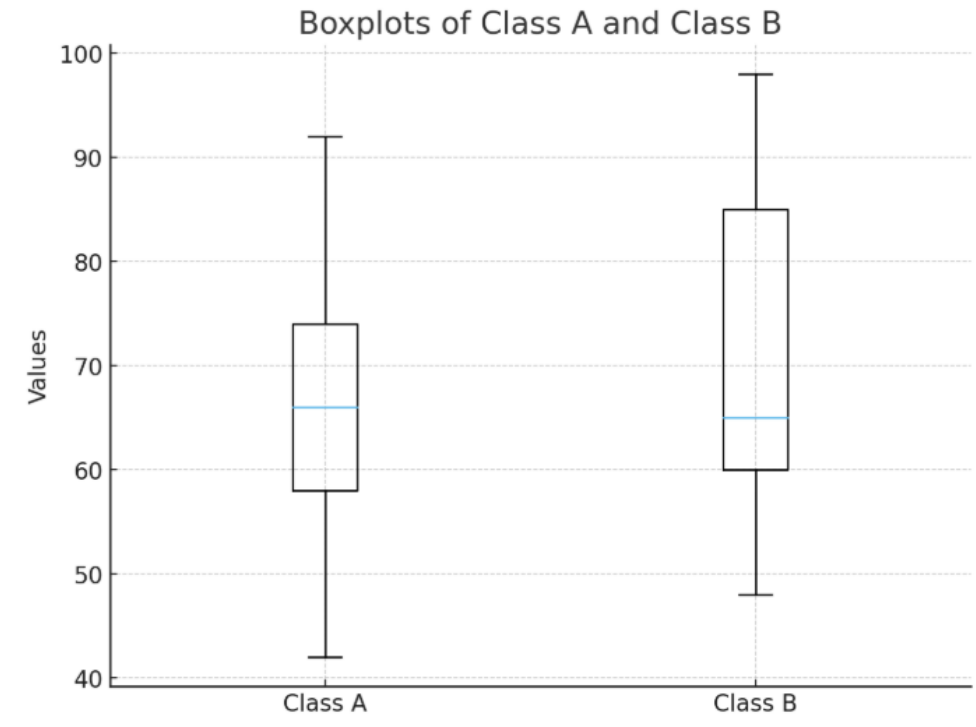
Class B: Median (65) is closer to Q1 (60) than Q3 (85). Big gap between Q1 and Q3 with median low suggests right-skewed (longer upper spread)

- Class A:
Q3 = 74. So ~25% of data lies above 74.
Since 75 is just 1 unit above Q3, a bit less than 25% (~20–25%) is above 75.

Class B:

Q3 = 85. So ~25% of data lies above 85.

Since $75 < 85$, that means more than 25% of data is above 75. A rough estimate could be around 30–35%.



Exercices

Exercise 3:

Test A: $\mu=70$, $\sigma=10$. Alice scored **85**.

Test B: $\mu=52$, $\sigma=6$. Ben scored **60**.

- Compute **z_Alice** and **z_Ben**. Who did better **relative to their class**?
- What raw score on Test A would match **z = 2**?
- What z-score corresponds to **x = 60** on Test A?

Exercices

Solution 3:

- $z_{\text{Alice}} = (85 - 70) / 10 = 1.5$ while $z_{\text{Ben}} = (60 - 52) / 6 \approx 1.33$
→ **Alice** did better.
- $x = \mu + 2\sigma = 70 + 2 * 10 = 90$
- $z = (60 - 70) / 10 = -1$

Exercices

Exercise 4:

Assume $X \sim N(50, 3.2^2)$

- Using the **empirical rule**, give a rough % of values between 46.8 and 53.2
- Compute the **z-score** for $x=41$
- Using a z-table estimate the percentage of values smaller or equal to 41.
- Interpret your answer in words.

Exercices

Solution 4:

- 68% (between 1sigma)
- $z=(41-50)/3.2= -9/3.2=-2.81$
- 0.25%
- Only 0.25% of the values are lower than 41

Exercices

Exercise 5:

Scores are $N(100, 15^2)$

- What raw score is the **84th percentile**?
- What raw score is the **2.5th percentile**?

Exercices

Solution 5:

- For the **84th percentile**: since 16% of the distribution is higher, we find the z where $P(Z < z) = 0.16$. That happens roughly at $z=1$ (0.99)

Therefore, if $z=(X-\mu)/\text{std} \rightarrow X= \mu+z*\text{std}=100+1*15=115$

- For the **2.5th percentile**: since 2.5% of the distribution is lower, we find the z where $P(Z < z) = 0.025$. That happens roughly at $z=-1.96$

Therefore, $X= \mu+z*\text{std}=100+-1.91*15=70.6$